§1 Recall on permutations.

- Length of a permutation $w \in S_n$ is
  - ~~Mess~~ $\min(\{p : \exists a_1, \dots, a_p : w = s_{a_1} \cdots s_{a_p}\})$
  - ~~#~~ $\#I(w)$, $I(w) = \{(i,j) \mid i<j, w(i) > w(j)\}$.
- Longest element $w_0 = \begin{pmatrix} 1 & \cdots & n \\ n & \cdots & 1 \end{pmatrix}$, w_0 is unique.

$$\ell(ww_0) = \ell(w_0 w) = \frac{n(n-1)}{2} - \ell(w).$$

- Reduced words

$$R(w) = \{(a_1, \dots, a_p) : p = \ell(w), s_{a_1} \cdots s_{a_p} = w\}.$$

- Exchange lemma:

$$(a_1, \dots, a_p), (b_1, \dots, b_p) \in R(w) \Rightarrow \exists i : (b_1, a_1, \dots, \hat{a_i}, \dots, a_p) \in R(w).$$

- Corollary If $w = s_{a_1} \cdots s_{a_p}$, $(a_1, \dots, a_p) \notin R(w)$

$$\Rightarrow w = s_{a_1} \cdots \hat{s_{a_i}} \cdots \hat{s_{a_j}} \cdots s_{a_p}.$$

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§2 Recall on divided differences

- Work over $\mathbb{C}(x_1, \dots, x_n)$.
- $\partial_i = \frac{1}{x_i - x_{i+1}} \cdot (1 - s_i)$
- ~~$\partial_i(f)$ is sym~~ $\partial_i(f \cdot g) = \partial_i(f) \cdot g + s_i(f)\, \partial_i(g)$
  ⇓ sym-linear.
- $\partial_i \curvearrowright \mathbb{C}[x_1, \dots, x_n]$.
- $\partial_i\left(\mathbb{C}(x_1, \dots, x_n)\right) \subseteq \left(\mathbb{C}(x_1, \dots, x_n)\right)^{s_i}$.
- $\partial_i(f) = 0 \iff f \in \mathbb{C}(x_1, \dots, x_n)^{s_i}$
  (in part. $\partial_i^2 = 0$).
- Rel-s: $\partial_i^2 = 0$, $\partial_i \partial_j = \partial_j \partial_i$, $|i-j| > 1$
  $\partial_i \partial_{i+1} \partial_i = \partial_{i+1} \partial_i \partial_{i+1}$.

- Let $w \in S_n$ then let $(a_1, \dots, a_p) \in R(w)$
  Let $\partial_w := \partial_{a_1} \cdots \partial_{a_p}$.
  Lemma ∂_w is well. def.
  Lemma If $(a_1, \dots, a_r)$ is non-reduced,
  $\partial_{a_1} \cdots \partial_{a_r} = 0$.

~~Cor~~ Cor. $\partial_u \partial_v = \begin{cases} \partial_{uv}, & \ell(uv) = \ell(u) + \ell(v) \\ 0, & \text{otherwise.} \end{cases}$

Example $\partial_{w_0} = \frac{1}{a_\delta} \sum_{w \in S_n} (-1)^{|w|} w$

$\partial_{w_0}\left(x^{\lambda+\delta}\right) = s_\lambda$.

§3| Recall on Schubert polynomials

- $\sigma_w := \partial_{w w_0}(X^\delta)$
- $\partial_r(\sigma_w) = \begin{cases} \sigma_{wr}, & \ell(wr) = \ell(w) - \ell(r) \\ 0, & \text{otherwise} \end{cases}$
- $\sigma_{w_0} = X^\delta$, $\sigma_{Id} = 1$.
- $\deg(\sigma_w) = \ell(w)$
- **Prop** (i) $\sigma_w = \sum_\alpha c_\alpha X^\alpha$ with $\alpha : \alpha_i \le \delta_i = n - i$

  (ii) $s_i \sigma_w = \sigma_w \iff \ell(ws_i) = \ell(w) + 1 \iff w(i) < w(i+1)$

  (iii) $\sigma_w \in \mathbb{Z}[x_1, \dots, x_r]$, where r is the last descent of w, i.e. $w(r+1) < w(r+2) < \dots$, but $w(r) > w(r+1)$.

- $\sigma_{s_i} = x_1 + \dots + x_i$.
- Stability $i : S_n \hookrightarrow S_m$

  $\Rightarrow \sigma_{i(w)} = \sigma_w \quad \forall w \in S_n$.

  $i : S_n \times S_m \longrightarrow S_{n+m}$, $(u, v) \longmapsto u \times v$.

  $\sigma_{u \times v} = \sigma_{u \times Id} \cdot \sigma_{Id \times v}$, with $\sigma_{u \times Id} = \sigma_u$

  $\Rightarrow$ Can define for $w \in S_\infty$.

§4 Basis properties

~~Def.~~

Thm $\{\sigma_w\}_{w\in S_n}$ is a $\mathbb{Z}$-basis for $\mathrm{Span}(\{X^\alpha\}_{\alpha\in\delta})$, ($n!$ dimensional).

Pf.

First we prove $\mathbb{Q}$-basis.

It is sufficient to prove independence. Suppose

$$\sum_{w\in S_n} a_w \sigma_w = 0.$$

Take ~~longest~~ largest a longest w s.t. $a_w \neq 0$. Denote it $\hat{w}$.

Then $0 = \partial_{\hat{w}}\left(\sum_{w\in S_n} a_w \sigma_w\right) = a_{\hat{w}}$

$$\partial_{\hat{w}}(\sigma_w) = \begin{cases} \sigma_{w\hat{w}^{-1}}, & \ell(w\hat{w}^{-1}) = \ell(w) - \ell(\hat{w}) \overset{\text{in our case}}{\geq} 0 \\ 0, & \text{otherwise} \end{cases}$$

$\ell(w\hat{w}^{-1}) = 0 \iff w = \hat{w}$

Therefore

$0 = a_{\hat{w}}$. But $a_{\hat{w}} \neq 0 \Rightarrow$ ↯.

~~Notice that~~

Assume $X^\alpha = \sum_{\substack{w \\ \ell(w)=|\alpha|}} a_w \cdot \sigma_w$

Then $a_w = \partial_w(X^\alpha) \in \mathbb{Z}[x_1,\dots,x_n] \cap \mathbb{Q} = \mathbb{Z}$.

∎

Recall

Let us denote $\Lambda_n = (\mathbb{Z}[x_1,\dots,x_n])^{S_n}$ → go over $\mathbb{Q}$.

Introduce grading by degree. $\Lambda_n = \bigoplus_{k\geq 0} \Lambda_n[k]$.

Then $\chi_{\Lambda_n}(q) = \sum_{k\geq 0} \dim(\Lambda_n[k])\, q^k$.

Then since $\Lambda_n = \mathbb{Z}[e_1,\dots,e_n]$

$$\chi_{\Lambda_n}(q) = \frac{1}{(1-q)\cdots(1-q^n)} = \frac{1}{\prod_{i=1}^{n}(1-q^i)}.$$

Prop. $\chi_{H_n}(q) = (n)_q! = \frac{(1-q^n)\cdots(1-q)}{(1-q)\cdots(1-q)} = \sum_{w\in S_n} q^{\ell(w)}$.

Pf.

$$\chi_{H_n}(q) = (1+\dots+q^{n-1})(1+\dots+q^{n-2})\cdots(1+q)\cdot 1 = (n)_q!$$

□

Notice that $\chi_{\mathbb{Z}[x_1,\dots,x_n]}(q) = \frac{1}{(1-q)^n}$.

Cor. $\chi_{\mathbb{Z}[x_1,\dots,x_n]}(q) = \chi_{H_n}(q)\cdot\chi_{\Lambda_n}(q)$.

Then As Λ_n module $\mathbb{Z}[x_1,\dots,x_n] \cong \bigoplus_{w\in S_n} \Lambda_n \cdot \sigma_w \cong \Lambda_n \otimes H_n$

Pf. Consider a map $\varphi : \left(\bigoplus_{w\in S_n} \Lambda_n\right) \longrightarrow \mathbb{Z}[x_1,\dots,x_n]$.

$$(a_w) \longmapsto \sum_w a_w \cdot \sigma_w$$

~~Thm~~

Cor $\{\sigma_w\}_{w \in S_\infty}$ is a $\mathbb{Z}$-basis for $\mathbb{Z}[x_1, x_2, \dots, x_n, \dots]$.

Prop. Let $S^{(n)} \subset S_\infty$ be $\{w : \text{last desc. of } w \text{ is} \le n\}$ $\Leftarrow \ell D(w)$

Then $\{\sigma_w\}_{w \in S^{(n)}}$ is a $\mathbb{Z}$ basis of $\mathbb{Z}[x_1, \dots, x_n]$

Pf. $\ell(\alpha) \le n$.

Let $x^\alpha = \sum_{w \in S_\infty} a_w \cdot \sigma_w = \sum_{w \in S^{(n)}} a_w \cdot \sigma_w + \sum_{w \in S_\infty \setminus S^{(n)}} a_w \cdot \sigma_w$.

Assume in the second sum some a_w is non-zero.

Let m be the maximal last descent for w s.t. $a_w \neq 0$.

$\Rightarrow \exists\, m > n$.

$$\partial_m(x^\alpha) = 0; \quad \partial_m(\sigma_w) = 0 \ \forall w \in S^{(n)}.$$

Then $0 = \sum_{\substack{w \in S_\infty \setminus S^{(n)} \\ LD(w) = m}} a_w \cdot \sigma_{w \cdot s_m}$ $\not\Leftarrow$ with indep. $\Rightarrow$

$\Rightarrow$ The second sum is empty. ∎

- φ is injective. Assume $\sum_w a_w \sigma_w = 0$

Take a longest $\hat{w}$ s.t. $a_w \neq 0$

$\Rightarrow \partial_{\hat{w}} \left(\sum_w a_w \sigma_w \right) = 0$

$\partial_{\hat{w}} \left(\sum_w a_w \sigma_w \right) = a_{\hat{w}}$

Since $\partial_{\hat{w}}$ is Λ_n-linear.

Therefore $(\Lambda_n \otimes H_n) \subset \mathbb{Q}[x_1, \dots x_n]$ (with grading preserved)

But $\chi_{\Lambda_n \otimes H_n}(q) = \chi_{\Lambda_n}(q) \cdot \chi_{H_n}(q) = \chi_{\mathbb{Q}[x_1, \dots x_n]}(q)$

Hence $\dim((\Lambda_n \otimes H_n)[k]) = \dim(\mathbb{Q}[x_1, \dots x_n][k]) \Rightarrow$

$\Rightarrow (\Lambda_n \otimes H_n)[k] = \mathbb{Q}[x_1, \dots x_n][k]$.

$\Rightarrow \Lambda_n \otimes H_n = \mathbb{Q}[x_1, \dots x_n]$.

∎

Lemma

Let $w \in S_n$, t_{ij} - be a transp. of i and j $(i<j)$.

Then $$\ell(wt_{ij}) = \begin{cases} \ell(w) + 2e_{ij}(w) + 1, & (i,j) \notin I(w) \\ \ell(w) - 2e_{ij}(w) - 1, & (i,j) \in I(w) \end{cases}$$

$$e_{ij}(w) = \#\{k : i<k<j, \ w(k) \text{ is between } w(i) \text{ and } w(j)\}.$$

Pf. Exercise / sketch ∎.

Lemma

If $f = \sum_{i=1}^{n} \alpha_i \cdot x_i$

Then $\partial_w (f \cdot g) = w(f)\,\partial_w(g) + \sum_{\substack{i<j \\ \ell(wt_{ij}) = \ell(w)-1}} (\alpha_i - \alpha_j)\,\partial_{wt_{ij}}(g)$.

Pf. Recall: $\partial_i(fg) = \partial_i(f)g + s_i(f)\partial_i(g)$

Write $w = s_{a_1} \cdots s_{a_p}$ - red. decomp.

$$\partial_{a_1} \cdots \partial_{a_p}(fg) = w(f)\partial_w(g) + \sum_{j=1}^{p} \partial_{a_j}\left(s_{a_{j+1}} \cdots s_{a_p}(f)\right) \partial_{a_1} \cdots \partial_{a_{j-1}} \partial_{a_{j+1}} \cdots \partial_{a_p}(g) =$$

$$= w(f)\partial_w(g) + \sum_{j=1}^{p} \partial_{a_j}\left(\sum_{k=1}^{n} \alpha_k x_{w_j^{(-1)}(k)}\right) \partial_{a_1} \cdots \widehat{\partial_{a_j}} \cdots \partial_{a_p}(g) =$$

$w = w_j^{(1)} s_{a_j} w_j^{(-)}$

$$= w(f)\partial_w(g) + \sum_{j=1}^{p} \left(\alpha_{w_j^{(-1)}(a_j)} - \alpha_{w_j^{(-1)}(a_j+1)}\right) \partial_{a_1} \cdots \widehat{\partial_{a_j}} \cdots \partial_{a_p}(g) =$$

$$= w(f)\partial_w(g) + \sum_{\substack{(i,j) \in I(w) \\ \ell(wt_{ij}) = \ell(w)-1}} (\alpha_i - \alpha_j)\,\partial_{wt_{ij}}(g) \quad (=)$$

$w = s_{a_1} \cdots s_{a_p}$

$s_{a_1} \cdots \widehat{s_{a_j}} \cdots s_{a_p} = w \cdot \left(s_{a_p} \cdots s_{a_{j+1}} s_{a_j} s_{a_{j+1}} \cdots s_{a_p}\right) = w\, t_{\left(w_j^{(+1)}(a_j),\, w_j^{(-1)}(a_j+1)\right)}$.

$$(=)\quad w(f)\partial_w(g) + \sum_{\substack{i<j \\ \ell(wt_{ij}) = \ell(w)-1}} (\alpha_i - \alpha_j)\,\partial_{wt_{ij}}(g).$$

□

Cor.

$$\partial_w(x_i g) = x_{w(i)}\,\partial_w(g) + \sum_{\substack{j:\, j>i \\ \ell(wt_{ij})=\ell(w)-1}} \partial_{wt_{ij}}(g) - \sum_{\substack{j:\, j<i \\ \ell(wt_{ji})=\ell(w)-1}} \partial_{wt_{ji}}(g).$$

Let $\ell(v) = \ell(w)+1$.

$$\partial_v(x_r \cdot \sigma_w) = \sum_{j:\, j>r} \underbrace{\delta_{vt_{rj},w}}_{=\delta_{v,wt_{rj}}} - \sum_{j:\, j<r} \underbrace{\delta_{vt_{jr},w}}_{=\delta_{v,wt_{jr}}}.$$

$\ell(wt_{rj})=\ell(w)-1$ $\Leftrightarrow$ $\ell(wt_{rj})=\ell(w)+1$; $\ell(wt_{jr})=\ell(w)-1$

Thm.

$$\Rightarrow \quad x_r \cdot \sigma_w = \sum_{v:\, \ell(v)=\ell(w)+1} \partial_v(x_r\sigma_w)\,\sigma_v = \sum_{t:\, \ell(wt)=\ell(w)+1} \varepsilon_w^r(t)\cdot \sigma_{wt}$$

With $$\varepsilon_w^r(t) = \begin{cases} 1, & t = t_{rj} \\ -1, & t = t_{jr} \\ 0, & \text{otherwise.} \end{cases}$$

Cor/Thm (Monk's formula)

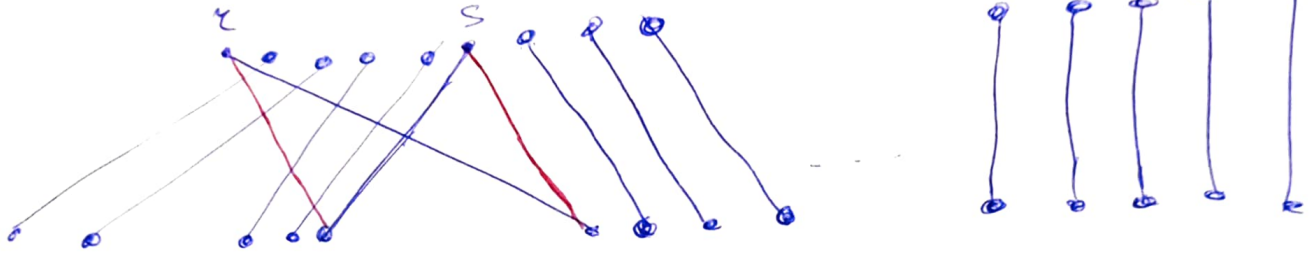
$$\sigma_{s_r}\cdot\sigma_w = \sum_{t:\, \ell(wt)=\ell(w)+1} \left(\sum_{i=1}^{r} \varepsilon_w^i(t)\right)\sigma_{wt} = \sum_{i\le r<j,\ \ell(wt_{ij})=\ell(w)+1} \sigma_{w\cdot t_{ij}}$$

∎

~~07~~/Thm

Let r be the last descent of w.

Let $s = \max\{i : w(i) < w(r)\} > r$.



Then: $\ell(wt_{rs}) = \ell(w) - 1$. and

$$X_r \sigma_{w \cdot t_{rs}} = \sigma_w - \sum_{q<r} \sigma_{w t_{rs} t_{qr}}.$$

$$\ell(w t_{rs} t_{qr}) = \ell(w)$$

$$\Rightarrow \sigma_w \in \mathbb{Z}_{\geq 0}[X_1, \dots, X_r].$$

$$\sigma_w = X_r \cdot \sigma_{w \cdot t_{rs}} + \sum_{q<r} \sigma_{w t_{rs} t_{qr}}.$$

$$\ell(w t_{rs} t_{qr}) = \ell(w)$$